Adaptive Candidate Routing for Distributed Occupant Recognition Across Multiple Building IoT-enabled Smart Spaces.

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Abstract

A multi-building camera network performing occupant identification and tracking often has to find out which building a seen individual's face belongs to in order to carry out recognition effectively. A centralized gallery of face embeddings facilitates this recognition task effectively, but it keeps sensitive biometric data in one place. The proposed work discusses a routing strategy that avoids the centralized reference repository, yet effectively recognizes and tracks the occupant. Each building publishes one Bloom filter built from locality-sensitive hashing (LSH) codes of its own occupants. The embedding extracted from an input photograph is converted into $\boldsymbol{L}$ LSH codes that are checked against all filters, and only the best-scoring buildings verify the face against their own reference embeddings. On 500 CASIA-WebFace identities in 10 buildings, averaged over 20 random building assignments, the default setting identifies 92.37% of input photographs correctly, against 95.47% for a centralized search with the same vote rule; an adaptive rule that asks extra buildings only when the ranking is unclear reaches 92.44% and contacts 2.54 buildings on average. The mean face and the accept angle come from reference photographs only; $\boldsymbol{K}$ and $\boldsymbol{P}$ were chosen after seeing test accuracy, so alternatives are reported. The real false-positive rate of the filters is 3.54% instead of the configured 1%; LSH code collisions cause the difference. When the same people are regrouped into more and smaller buildings, the share of input photographs whose true building is in the shortlist falls from 94.8% at 10 buildings to 87.0% at 100. Someone who holds the filters can tell which building a

person belongs to (AUC 0.93). Costs are computed, not measured; encryption and unknown visitors are not evaluated.

Keywords: Occupant identification and tracking, Smart Spaces, Locality-sensitive hashing, Bloom filter, Privacy in data analytics

1 Introduction

When a face is seen at one building of a campus or company, the system often has to find out who the person is and where that person is registered. The usual answer is one central gallery with the face embeddings of everyone. This is simple and accurate, but it stores sensitive biometric data in a single place, which is an attractive target and may conflict with data-minimization rules. The other extreme is a gallery in each building and a request to every building for every input photograph, which exposes every input photograph to every building.

We study the step between these two: a routing step that decides which few buildings should be asked. Each building publishes one Bloom filter [1] built from LSH codes of its own occupants. The input embedding is checked against all filters, and only the best-scoring buildings receive it. Reference embeddings are not published. The idea of publishing a compact summary of what each site holds, and checking the summaries before contacting a site, is used in web cache sharing [2]. Face matching is approximate, so the summary cannot hold exact keys; it holds LSH codes instead.

Scope. This paper is about the routing step. We measure how accurate it is, how many buildings it contacts, how many false positives the filters give, and how it behaves when there are more buildings. All evaluated input photographs depict enrolled people; identifying unknown visitors is left for future work. We do not implement or test any cryptographic protection. We assume that messages between buildings use a standard authenticated, encrypted channel, and Section 3 states what is and is not protected. Storage and communication costs are computed from message sizes; no distributed run was measured.

Our contributions are:

- a routing design with one Bloom filter of LSH codes per building, and a way to choose the code length K and the number of codes L from measured angle distributions;
- measurements on 500 identities and 20 building assignments: accuracy against a centralized search, the gain from multi-probe hashing, and a real false-positive rate of 3.54% that we trace to collisions between LSH codes;
- a rule that asks extra buildings only when the two best scores are close, which is as accurate as asking all shortlisted buildings and contacts about half as many;
- an evaluation with up to 100 buildings, an analytic cost comparison, and a measurement of what the published filters reveal about who is in a building.

2 Related Work

Existing studies of indoor occupant tracking use cameras [3] or RF signals [4], and fit the broader view of cyberphysical spaces, where physical environments and computational systems are considered together [5]. Across multi-building smart spaces, computational efficiency and privacy preservation remain the two main challenges, especially with camera-based sensing [6]. Facial biometrics suit identification in IoT-enabled smart spaces [7] because they carry distinctive identity information and can be captured remotely, without the person stopping, cooperating, or presenting a credential. Mohan and Menon [3] model occupant identification and tracking across buildings as a distributed state transition system; we study how an input embedding can be routed to the building that holds the person.

Template protection and Bloom filters. Jain et al. [8] survey the protection of stored biometric templates, and Rathgeb et al. [9] apply Bloom filters to iris templates. Closest to our work, Marzani et al. [10] turn face features into identifiers with a fuzzy extractor and store them in homomorphically encrypted Bloom filters to count people across locations. Their goal is counting by exact identifier membership; ours is routing an input embedding by approximate similarity (LSH codes with multi-probe) to the building that holds the person.

Routing in distributed search. Summary Cache [2] lets web proxies exchange Bloom filters of their contents and contact only matching proxies; our routing step follows this idea with LSH codes in place of exact keys. For vector data, Gottesbüren et al. [11] study clustering- and LSH-based methods that send a search request only to shards likely to hold its neighbours. Semord [12] places similar vectors close together in a decentralized overlay using a learned hash; its peers exchange summaries that include representative vectors, and it is evaluated on generic benchmarks, not on faces or with a privacy goal. FRAG [13] supports nearest-neighbour search over encrypted vectors held by mutually distrusting parties, but sends each input to every node and has no routing step. These systems route by vector placement, centroids or learned structure, which requires moving data or sharing vector summaries. We instead let each building publish a small Bloom filter of LSH codes and keep its reference embeddings. To our knowledge, the combination of LSH codes and per-building Bloom filters has not been measured for this task before.

3 System and Threat Model

We consider a multi-building occupant tracking environment in which each building stores reference face embeddings for $N = 50$ enrolled occupants, obtained from $R = 20$ photographs per occupant. An input photograph contains the face of the person to be identified. We assume that the person depicted in each input photograph is enrolled in exactly one building. Each building constructs a Bloom filter from the Locality-Sensitive Hashing (LSH) codes of its enrolled occupants. The system uses these filters to shortlist candidate buildings, which perform local face verification using their stored reference embeddings. Reference embeddings are not exchanged between buildings during routing or verification, although input embeddings are sent to shortlisted buildings. We assume that buildings execute these operations correctly but may attempt

to infer sensitive information from received data. Published Bloom filters, public hash parameters, and input photographs may reveal building membership, as evaluated in Section 6.6. Authenticated and encrypted communication is assumed but not implemented or evaluated. Candidate-face guessing, external information, repeated input photographs, colluding buildings, malicious behavior, and traffic analysis are outside the scope of this study.

4 Methodology

The method has two phases: each building builds its Bloom filter once (Algorithm 1), and every input photograph is then routed and verified (Algorithm 2). We first define the quantities that the algorithms use.

Embedding and LSH codes.

SCRFD-10GF [14] detects each face, and pretrained ArcFace (InsightFace `buffalo_l`) [15] produces a 512-dimensional embedding. For input photograph I_p with raw embedding $\tilde{\mathbf{x}}_p$, we first unit-normalize and then subtract the reference-only population mean $\bar{\mathbf{x}}$ and renormalize:

$$\mathbf{x}_p = \frac{\tilde{\mathbf{x}}_p}{\|\tilde{\mathbf{x}}_p\|_2}, \qquad \mathbf{p} = \frac{\mathbf{x}_p - \bar{\mathbf{x}}}{\|\mathbf{x}_p - \bar{\mathbf{x}}\|_2}. \tag{1}$$

The same mean and transformation are used for enrollment and identification. Let K be the number of bits per code and L the number of slots. Using KL fixed-seed Gaussian hyperplanes $\mathbf{h}_{\ell,i}$, define $b_{\ell,i} = \mathbb{1}[\mathbf{p} \cdot \mathbf{h}_{\ell,i} \geq 0]$ and the slot-tagged item

$$c_\ell = \sum_{i=0}^{K-1} b_{\ell,i} 2^i, \qquad s_\ell = \ell 2^K + c_\ell. \tag{2}$$

For angle θ between two unit embeddings, a bit matches with probability $p_{\text{bit}} = 1 - \theta/180$ [16].

Parameter selection.

Averaging p_{bit}^K over reference-only genuine and impostor pairs gives $p_{\text{code}}^{(G)}$ and $p_{\text{code}}^{(I)}$. Assuming independent slots,

$$\text{Recall} = 1 - (1 - p_{\text{code}}^{(G)})^L, \quad \text{FPR}_{\text{imp}} = 1 - (1 - p_{\text{code}}^{(I)})^L. \tag{3}$$

For target recall 0.90, set $L = \left\lceil \ln(0.10) / \ln(1 - p_{\text{code}}^{(G)}) \right\rceil$; sweep $K = 4, \ldots, 15$, reject predicted $\text{FPR}_{\text{imp}} > 0.10$, and choose the smallest KL. This selects $(10, 88)$ analytically. The main experiments use $(11, 122)$ because it performed better on test accuracy; this test-informed choice and the alternative are reported in Section 6.1.

Per-building Bloom filters.

Each building publishes a Bloom filter but retains its reference embeddings. For occupant u in building b, the normalized mean of the R preprocessed reference embeddings forms centroid $\mathbf{z}_{b,u} = \text{Normalize}(R^{-1} \sum_{r=1}^{R} \mathbf{e}_{b,u,r})$, which is encoded into L items using Equation 2. With $n = NL$ inserted items and target false-positive probability $f^\star$, the filter size and number of hash positions are

$$m = \left\lceil \frac{-n \ln f^\star}{(\ln 2)^2} \right\rceil, \qquad \kappa = \text{round}\Big(\frac{m}{n} \ln 2\Big). \tag{4}$$

For $N = 50$, $R = 20$, $L = 122$, and $f^\star = 0.01$, this gives $n = 6100$, $m = 58{,}469$, and $\kappa = 7$ (about 7.3 KB). We use Kirsch–Mitzenmacher double hashing [17]: $h_1 = \text{SM64}(s \oplus (2\sigma+1))$, $h_2 = \text{SM64}(s \oplus (2\sigma+2)) \vee 1$, and $p_j = ((h_1 + j h_2) \bmod 2^{64}) \bmod m$ for $j = 0, \ldots, \kappa - 1$, where σ is a shared salt and SM64 is the SplitMix64 finalizer [18]. A check is positive when all κ positions are set.

Algorithm 1 Building the published Bloom filter at building b

Input: Reference photographs $I_{b,u,r}$ of the N occupants of building b (R per occupant);
population mean $\bar{\mathbf{x}}$; shared hyperplanes $\mathcal{H}$ and salt σ; parameters $K, L, f^\star$.
Output: Published Bloom filter F_b; private gallery $\mathcal{E}_b$.

$n \leftarrow NL$; derive m and κ ▷ Equation 4
$F_b \leftarrow$ all-zero bit array of length m
for $u = 1$ to N **do**
  for $r = 1$ to R **do**
    $\mathbf{e}_{b,u,r} \leftarrow$ preprocessed embedding of $I_{b,u,r}$ ▷ Equation 1
  end for
  $\mathbf{z}_{b,u} \leftarrow \text{Normalize}\big(R^{-1} \sum_r \mathbf{e}_{b,u,r}\big)$ ▷ occupant centroid
  for $\ell = 0$ to $L - 1$ **do**
    $s_\ell \leftarrow$ slot-tagged item of $\mathbf{z}_{b,u}$ ▷ Equation 2
    set the κ bits $p_0, \ldots, p_{\kappa-1}$ of item s_ℓ in F_b ▷ double hashing
  end for
end for
$\mathcal{E}_b \leftarrow \{\mathbf{e}_{b,u,r}\}$, kept private; publish F_b

Routing, verification, and policy.

For each slot, multi-probe routing [19] checks the original code and P single-bit variants obtained by flipping the bits with the smallest margins $|\mathbf{p} \cdot \mathbf{h}_{\ell,i}|$:

$$s'_{\ell,j} = \ell 2^K + \big(c_\ell \text{ XOR } 2^{w_{\ell,j}}\big), \quad j = 1, \ldots, P, \tag{5}$$

where $w_{\ell,j}$ is the index of the j-th smallest-margin bit. Building b receives score A_b, the number of slots with at least one positive check; the shortlist $\mathcal{C}$ contains the top $\min(S, B)$ buildings by score (ties by building index). At a shortlisted building, each

reference embedding votes for its occupant when its angle to the input embedding is at most $\tau = 80.43°$; the local winner is the occupant with most votes (ties by smallest reference angle), and acceptance requires at least $V = 12$ of $R = 20$ votes. The adaptive policy verifies all shortlisted buildings when $g = A_{(1)} - A_{(2)} \leq g_0$ and selects by vote count, minimum reference angle, then routing rank; otherwise it stops at the first local acceptance in routing order. If none is accepted, it abstains. The default all-shortlist rule used in the multi-probe comparison instead compares votes across all occupants in $\mathcal{C}$, breaking ties by minimum reference angle.

Algorithm 2 Adaptive candidate routing and recognition

Input: Input photograph I_p of a person to be identified;
published filters $\{F_b\}_{b=1}^{B}$; private galleries $\{\mathcal{E}_b\}_{b=1}^{B}$ held by the buildings;
mean $\bar{\mathbf{x}}$, hyperplanes $\mathcal{H}$, parameters K, L, P, S, V, τ, g_0.
Output: Identified occupant or ABSTAIN.

$\mathbf{p} \leftarrow$ preprocessed embedding of I_p ▷ Equation 1
for $\ell = 0$ to $L - 1$ **do**
  compute s_ℓ and the P probe items $s'_{\ell,1}, \ldots, s'_{\ell,P}$ ▷ Equations 2 and 5
end for
for $b = 1$ to B **do**
  check the original and probe items of every slot against F_b
  $A_b \leftarrow$ number of slots with at least one positive check ▷ each slot counted once
end for
$\mathcal{C} \leftarrow$ top $\min(S, B)$ buildings by decreasing A_b ▷ ties by building index
$g \leftarrow A_{(1)} - A_{(2)}$ if $B \geq 2$, else $g \leftarrow +\infty$
if $g \leq g_0$ **then** ▷ ranking unclear: verify all shortlisted buildings
  for all $b \in \mathcal{C}$ **do**
    $(u_b, v_b) \leftarrow \textsc{LocalVerify}(b, \mathbf{p})$
  end for
  $\mathcal{A} \leftarrow \{b \in \mathcal{C} : v_b \geq V\}$
  if $\mathcal{A} = \emptyset$ **then**
    return ABSTAIN
  end if
  return u_b for the $b \in \mathcal{A}$ with most votes ▷ ties: smallest angle, then routing rank
else ▷ ranking clear: sequential first-match
  for all $b \in \mathcal{C}$ in routing order **do**
    $(u_b, v_b) \leftarrow \textsc{LocalVerify}(b, \mathbf{p})$
    if $v_b \geq V$ **then**
      return u_b
    end if
  end for
  return ABSTAIN
end if

function $\textsc{LocalVerify}(b, \mathbf{p})$ ▷ runs at building b on its private gallery
  for all occupant u of building b **do**
    $v_u \leftarrow |\{r : \angle(\mathbf{p}, \mathbf{e}_{b,u,r}) \leq \tau\}|$
  end for
  return $u^\star = \arg\max_u v_u$ (ties by smallest reference angle) and $v_{u^\star}$
end function

5 Experimental Setup

We use 500 identities of CASIA-WebFace [20], spread over $B = 10$ buildings of 50 people. Each person has 20 enrolled reference photographs and 20 held-out test photographs, giving 20,000 embeddings. Each of the 10,000 test photographs is processed by routing its input embedding over all B buildings. The evaluation is *closed-set*: every input photograph depicts an enrolled person. We use $K = 11$, $L = 122$, $P = 3$, $S = 5$, $V = 12$ and $f^\star = 0.01$.

Use of test photographs. The mean face, the accept angle and the angle statistics behind K and L are estimated from reference photographs only: the mean face is the mean of the 10,000 reference embeddings, genuine pairs are two reference photographs of the same person, and impostor pairs are reference photographs of different people. This gives $\tau = 80.43°$. An earlier version used test photographs for the mean face and τ (80.54°); with reference-only estimates, correct-identification rates changed by at most 0.06 points, and misidentification rates fell by 0.1 to 0.15 points because τ is lower. Since $K = 11$ and $P = 3$ were chosen after looking at test accuracy, we also report $K = 10$, the model's choice, and the full sweep over P (Section 6.1).

Seeds. Each result is repeated for 20 seeds. A seed changes three things together: the assignment of the 500 identities to buildings of 50, the LSH hyperplanes, and the Bloom filter salt. The embeddings, mean face and accept angle stay fixed, so the centralized baseline is a single number. We report the mean over seeds with a 95% confidence half-width ($\pm$, Student-t, 19 degrees of freedom) and, for comparisons, the paired difference over seeds. For $B = 2$ and $B = 5$, we regroup the identities of the first two and first five buildings of the original layout into buildings of 50. A single run with the original assignment and hyperplane seed 42 gave 92.32% at $P = 3$ with the default rule, 0.05 points below the 20-seed mean; 14 of the 20 seeds reach that value.

Metrics. Each input photograph results in exactly one of four outcomes: *correct*; *misidentified* (a wrong person is accepted, the most serious error); *abstained-reject* (the true building was shortlisted but nobody reached V votes); and *abstained-routing* (the true building was not shortlisted and nobody reached V votes). The centralized baseline gives 95.47% correct, 1.41% misidentified and 3.12% abstained.

6 Results

6.1 Choice of K and L

The analytic model uses only the angle arrays, so it has no seed. Using the mean angle first gives $p^{(G)}_{\text{code}} = 0.0115$ at $K = 11$, against the correct array average of 0.0187, which is about 39% too low. The model picks $K = 10$, $L = 88$. In the full pipeline at $P = 3$ with the default rule, $K = 11$, $L = 122$ is more accurate than $K = 10$ by 0.52 ± 0.13 points (paired) and better in 20 of 20 seeds: correct 92.37% against 91.85%, mainly because routing failures fall from 4.88% to 4.46%. Both results are reported; we use $K = 11$ in the rest of the paper, and we note that this choice was made after seeing test accuracy.

6.2 Multi-Probe and Decentralized versus Centralized Identification

Table 1 varies P at $B = 10$ with the default rule. Without probing, $8.07 \pm 0.11\%$ of input photographs are lost to routing misses. Multi-probe with $P = 3$ reduces this to $4.46 \pm 0.08\%$ (a 45% relative reduction), raises correct identification from 88.28% to 92.37% (a paired gain of 4.09 ± 0.15 points, positive in 20 of 20 seeds) and lowers misidentification from 1.91% to 1.42%. The gains get smaller as P grows, while the number of lookups grows linearly, so we use $P = 3$ as a compromise on cost, not as the best possible value.

Table 1 Multi-probe sweep at $B = 10$ ($K = 11$, $L = 122$, default rule). Results are reported as the mean over 20 seeds ($\pm$ 95% CI), in %. The number of items looked up per filter and input photograph is $L(1 + P)$.

P	Correct	Misid.	Abst.-reject	Abst.-routing	Items
0	88.28 ± 0.13	1.91 ± 0.04	1.74 ± 0.03	8.07 ± 0.11	122
1	90.92 ± 0.09	1.61 ± 0.04	1.71 ± 0.04	5.76 ± 0.11	244
2	91.90 ± 0.09	1.45 ± 0.04	1.73 ± 0.05	4.92 ± 0.10	366
3	92.37 ± 0.08	1.42 ± 0.03	1.75 ± 0.04	4.46 ± 0.08	488
5	92.80 ± 0.07	1.38 ± 0.03	1.77 ± 0.05	4.05 ± 0.07	732
7	92.89 ± 0.07	1.38 ± 0.03	1.76 ± 0.05	3.97 ± 0.08	976

We now compare the decentralized system with the centralized baseline. At $B = 2$ and $B = 5$ the shortlist of $S = 5$ covers every building, so routing cannot fail and the two systems give the same result in every seed (95.75% correct at $B = 2$ and 95.64% at $B = 5$). At $B = 10$ the decentralized system is 3.10 ± 0.08 points behind the baseline, and 4.46 points of input photographs are routing failures. Before multi-probe the gap was 7.19 points, so multi-probe closes 57% of it. Misidentification (1.42% against 1.41%) hardly changes, so the loss comes from missed matches and not from more wrong accepts.

6.3 Effective False-Positive Rate

The 1% target describes the bit array for items that are independent of the stored ones. We measured the rate the system really has by building all ten filters and testing the enrolled items of the *other nine* buildings against each one. The real per-item rate is $3.54 \pm 0.02\%$ over the 20 seeds. When we compare the integer codes directly, without a Bloom array, $2.61 \pm 0.02\%$ of the items of other buildings already equal an enrolled item in the same slot. This is close to the 2.41% expected if the codes of 50 occupants were evenly spread over $2^{11} = 2048$ values, namely $1 - (1 - 2^{-K})^N$. The Bloom array alone, tested on random integers that were never inserted, gives $0.94 \pm 0.01\%$ at 51.3% fill, close to its design rate. The extra false positives therefore come from the LSH codes, not from the Bloom array.

A larger K gives more possible codes and lowers the real false-positive rate, but L must grow to keep recall at 0.90. At $K = 13$, $L = 230$ each filter grows from 7.3 KB

to about 13.8 KB and the real rate falls to 1.67%. With $P = 6$ it is 0.21 ± 0.08 points more accurate than the best $K = 11$ setting ($P = 7$, 92.89%; paired, better in 19 of 20 seeds) and 0.73 ± 0.11 points more accurate than the default (93.10% against 92.37%; better in 20 of 20 seeds), but it needs 1,610 lookups per filter and input photograph against 488 or 976.

6.4 Verification Policies

Table 2 compares the policies of Section 4 on the same 10,000 input photographs. These results use the policies' own rule for choosing among buildings (best occupant per building, ties by rank order), so verify-all gives 92.34% here and not the 92.37% of the default rule in Table 1. The two differ only when two shortlisted buildings return the same number of votes, which happens for 72.5 of 10,000 input photographs on average. Each input embedding is scored against all ten building filters, including the filter for the person's registered building, so there is no separate local-recognition stage. Sequential first-match raises the misidentification rate to 2.17%, because a wrong building ranked above the true one sometimes confirms first. Asking all five buildings lowers it to 1.45% and gains 0.71 ± 0.03 points, but it contacts 5 buildings per input photograph. The adaptive rule with $g_0 = 5$ is slightly more accurate than asking all five ($+0.10 \pm 0.02$ points, paired, better in 20 of 20 seeds) while contacting 2.54 buildings on average; 37.6% of the input photographs ask all five. All three policies abstain on exactly the same input photographs, because acceptance depends only on whether some shortlisted building reaches V votes.

Table 2 Verification policies at $B = 10$ using the policy rule. Results are reported as the mean over 20 seeds (± 95% CI). Contacting every building would involve 10 buildings.

Policy	Correct (%)	Misid. (%)	Buildings asked	Asked all five
Sequential first-match	91.62 ± 0.08	2.17 ± 0.04	1.50	–
Adaptive, $g_0 = 5$	$\mathbf{92.44 \pm 0.09}$	1.35 ± 0.04	2.54	37.6%
Verify all five	92.34 ± 0.09	1.45 ± 0.04	5.00	100%

We repeated the comparison at two and five buildings (the same people regrouped into buildings of 50 for each seed; the shortlist is every building). Table 3 shows that the adaptive rule helps wherever ranking errors occur. At $B = 2$ the gain over sequential first-match is small (0.07 ± 0.03 points, better in 14 of 20 seeds), because with two buildings a wrong building rarely outranks the true one. At $B = 5$ the adaptive rule gains 0.64 ± 0.05 points over sequential first-match (better in 20 of 20 seeds), is $+0.01 \pm 0.02$ points above asking all five buildings, and contacts 2.29 rather than 5 buildings on average (54% fewer). Misidentifications fall from 1.37% (sequential) to 0.74% (adaptive). Verify-all is within 0.01 points of the centralized search at $B = 5$ (95.63% against 95.64%).

Table 3 Verification policies at $B = 2$ and $B = 5$ using the policy rule. Results are reported as the mean over 20 seeds (± 95% CI). Verify-all asks every building.

	$B = 2$			$B = 5$		
Policy	Correct (%)	Asked	All	Correct (%)	Asked	All
Sequential first-match	95.66 ± 0.04	1.09	–	95.01 ± 0.06	1.35	–
Adaptive, $g_0 = 5$	95.73 ± 0.02	1.20	18.0%	$\mathbf{95.64 \pm 0.02}$	2.29	31.4%
Verify all	95.75 ± 0.01	2.00	100.0%	95.63 ± 0.01	5.00	100.0%

6.5 Cost

The cost figures are computed from message sizes and from the measured number of buildings asked; they are not measurements. We did not measure latency, throughput or traffic in a distributed run, because all calls were in-process. The input embedding has 512 single-precision values, which is 2,048 bytes. Routing replaces a 20.5 MB gallery in one place with ten published filters of 7.3 KB each (the buildings keep 2.05 MB of private embeddings each). Processing one input photograph costs 3.1 KB with sequential asking (1.50 buildings), 5.2 KB with the adaptive rule (2.54 buildings) and 10.2 KB when all five shortlisted buildings are asked, against 20.5 KB when all ten buildings are asked and 2.0 KB for a central gallery. Each input photograph still costs $B \cdot L(1 + P) = 4{,}880$ filter lookups, a cost that grows linearly with B.

6.6 What the Filters Reveal

We measured the simplest attack. Someone who holds all published filters and the public hash parameters scores a photograph against every filter, in the same way as routing does. With $P = 3$, this score separates the occupants of a building from everyone else with an AUC of 0.93 (0.88 without probing), averaged over buildings and 10 seeds, and the highest-scoring building is the person's own building for 79.7% of the photographs (66.7% without probing). The filters therefore leak which building a person belongs to. This is not a way to recover a face, but it is a real leak, and it grows with the accuracy of the routing itself. We did not test attacks that use many input photographs, candidate guessing or outside information, so we make no claim of biometric privacy.

6.7 More Buildings

The corpus has only 500 identities, so a larger network can only be built with smaller buildings. We regroup the same 500 people into $B = 10$, 25, 50 and 100 buildings of 50, 20, 10 and 5 people. Each filter is sized for its own number of occupants, so the total size of all filters stays the same (about 73 KB). We use 2,000 input photographs per seed (4 photographs per person) and 10 seeds. Table 4 shows that the share of input photographs whose true building is in the shortlist falls from 94.8% to 87.0%, and correct identification falls from 92.5% to 86.7% (adaptive policy), while the centralized search stays at 95.5%. The adaptive policy still asks about 2.5 buildings. The real false-positive rate of a filter falls with building size, from 3.54% to 1.27%, so the loss

comes from having to pick 5 buildings out of more of them, not from more false hits per filter. Lookups per input photograph grow in proportion to B (from 4,880 to 48,800). These results are for smaller buildings and a fixed set of people; they do not replace a test with a larger population.

Table 4 Growth in the number of buildings (same 500 people, $K = 11$, $L = 122$, $P = 3$, $S = 5$): mean over 10 seeds, in %. Recall is the share of input photographs whose true building is in the shortlist.

B	people/bldg	recall	correct (adaptive)	correct (all five)	real FPR	filter size
10	50	94.8 ± 0.3	92.5 ± 0.3	92.4 ± 0.3	3.54%	7.3 KB
25	20	90.7 ± 0.4	89.8 ± 0.5	89.7 ± 0.5	2.04%	2.9 KB
50	10	89.1 ± 0.2	88.6 ± 0.3	88.6 ± 0.3	1.52%	1.5 KB
100	5	87.0 ± 0.4	86.7 ± 0.4	86.7 ± 0.4	1.27%	0.7 KB

7 Limitations and Conclusion

Limitations. All results use the same 500 identities and photographs from one public dataset; the intervals describe the effect of building assignments and hash randomness, not a different population. The values $K = 11$ and $P = 3$ were chosen after seeing test accuracy, and we report the alternatives next to them. CASIA-WebFace photographs come from the web and not from surveillance cameras, so the absolute accuracy is probably higher than it would be in practice. The ArcFace model was pretrained on a large public face collection, and we did not check how much it overlaps with CASIA-WebFace. The largest network has 100 small buildings. The evaluation is closed-set, so we did not measure what happens when a person who is enrolled nowhere, such as a visitor, is seen. The vote rule (τ and $V = 12$) was chosen for enrolled people, and a first test with held-out people showed that it can accept a noticeable share of them, so the acceptance rule has to be studied and tightened before the system is used with unknown visitors. Storage and communication costs are computed from message sizes; we did not measure latency or traffic in a distributed run. We did not implement or evaluate the encrypted channel, the visit-length storage of templates or attacks on the filters beyond the membership test of Section 6.6. A Bloom filter cannot delete items, so a change in enrollment needs a rebuild.

Conclusion. Per-building Bloom filters of LSH codes can route input embeddings to the right building without a central gallery. With multi-probe hashing the default setting identifies 92.37% of input photographs correctly at 10 buildings against 95.47% for a centralized search, and the adaptive rule reaches 92.44% while contacting about 2.54 buildings. The cost is lost recall that grows with the number of buildings, and a false-positive rate that LSH collisions raise well above the Bloom filter's design value. The privacy conclusions are limited to what was evaluated: the design avoids publishing reference embeddings and a central gallery, and it leaks which building a person

belongs to (AUC 0.93); encryption, stronger attacks and template storage during visits were not evaluated. The next steps are a larger and more varied population, the identification of unknown visitors with a stricter acceptance rule, and measurements in a distributed deployment.

Acknowledgement. The authors gratefully acknowledge Sri Mata Amritanandamayi Devi (Amma), Chancellor, Amrita Vishwa Vidyapeetham, for her inspiration and for providing financial support for the Article Processing Charges (APC) of this publication.

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